Group H - Proposal

This is a strong and well-motivated proposal, with a particularly interesting attempt to distinguish what the data directly observe from what you actually want to measure.

The main point to clarify is the definition of "failure." A station with no bikes may be perfectly operational but unavailable to a user, while a station with bikes may have a technical problem. Similarly, define more precisely what makes a station "isolated" and what counts as a viable alternative.

Be cautious with the distinction between "broken" and "unused." The bike-counter data can provide useful context about cycling activity, but they cannot by themselves establish that a particular Vélib' station is broken.

Your data collection strategy also raises an interesting temporal sampling question. Why collect every five minutes? What changes can you actually observe at that frequency, given the 20-30 minute delay in the feed? Make the sampling interval and reporting latency part of your methodological discussion, rather than treating them only as technical limitations. Also make sure that collecting continuously for ten weeks is realistically manageable and that you have a way to detect missing periods.

Finally, the scope is ambitious: station availability, neighbouring stations, weather, commuting patterns, holidays, cycling counters, geographic comparisons, and storage-format comparisons could easily become several projects. I would strongly recommend identifying the core analysis and considering whether one or two of these components could be dropped or simplified.

Overall, there is a lot of potential here. The project will become stronger if you narrow the scope and make the definitions, sampling strategy, and limits of what you can infer explicit.